



Inspiring Excellence

Department of Mathematics and Natural Sciences
MAT110 : Differential Equations and Coordinate Geometry

Assignment-1

Faculty : Tapan biswas(TBW)

Make a Front Page by yourself, mentioning your name, ID, and section.(Compulsory)

Answer All Questions

Total=15

1. Evaluate:

[2]

$$\lim_{x \rightarrow 0} \left(\frac{\tan x - x}{x^3} \right)$$

2. Evaluate:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} \right)^{2x}$$

3. Check continuity & differentiability of

[3]

$$f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0 \end{cases}$$

at $x = 0$.

4. Let $f(x) = \cos(5x)$. Using the basic rules of differentiation, find $f'(x)$.

[2]

5. Find equation of tangent and normal to the curve $x^2y + y^2 + x - 2y = 0$ at the point $(1, 1)$.

[2]

6. Find $\frac{dy}{dx}$ of the following functions:

[3]

(a) $y = (\ln x)^{\sin x}$

(b) $y = \frac{(x^2 + 2)^4 e^{3x}}{(x^2 + 1)^3}$

7. Check continuity & differentiability of $f(x) = |x + 1| + |x - 2|$ at $x = -1$ & $x = 2$

[3].

Best of Luck.